

Adventist University of Central Africa (AUCA)

Master of Science in Big Data Analytics

An Introduction to Statistical Learning

Chapter 4 – Classification

Solutions to Questions 2, 7, and 12

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Course

Statistical / Machine Learning – Big Data Analytics

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1 Question 2

Problem Statement

It was stated in the text that classifying an observation to the class for which (4.17) is largest is equivalent to classifying an observation to the class for which (4.18) is largest. Prove that this is the case.

Solution

We are given that the observations in class k follow a normal distribution:

$$X \mid Y = k \sim N(\mu_k, \sigma^2)$$

The Bayes classifier assigns an observation x to the class k for which the posterior probability $p_k(x) = P(Y = k \mid X = x)$ is largest.

From equation (4.17):

$$p_k(x) = \frac{\pi_k \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x - \mu_k)^2}{2\sigma^2}\right)}{\sum_l \pi_l \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x - \mu_l)^2}{2\sigma^2}\right)}$$

The denominator is the same for every class k . Therefore, maximizing $p_k(x)$ is equivalent to maximizing its numerator:

$$\pi_k \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x - \mu_k)^2}{2\sigma^2}\right)$$

Because the logarithm is a monotonically increasing function, taking the logarithm does not change which class maximizes the expression. Thus, we maximize:

$$\log(\pi_k) - \frac{1}{2} \log(2\pi\sigma^2) - \frac{(x - \mu_k)^2}{2\sigma^2}$$

Expand the squared term:

$$(x - \mu_k)^2 = x^2 - 2x\mu_k + \mu_k^2$$

Substituting gives:

$$\log(\pi_k) - \frac{1}{2} \log(2\pi\sigma^2) - \frac{x^2}{2\sigma^2} + \frac{x\mu_k}{\sigma^2} - \frac{\mu_k^2}{2\sigma^2}$$

Rearranging:

$$\frac{x\mu_k}{\sigma^2} - \frac{\mu_k^2}{2\sigma^2} + \log(\pi_k) - \frac{x^2}{2\sigma^2} - \frac{1}{2} \log(2\pi\sigma^2)$$

The terms $-\frac{x^2}{2\sigma^2}$ and $-\frac{1}{2} \log(2\pi\sigma^2)$ do not depend on k . They are therefore the same for all classes and can be ignored when determining which class gives the maximum.

We are left with:

$$\delta_k(x) = \frac{x\mu_k}{\sigma^2} - \frac{\mu_k^2}{2\sigma^2} + \log(\pi_k)$$

This is exactly equation (4.18):

$$\delta_k(x) = x \cdot \frac{\mu_k}{\sigma^2} - \frac{\mu_k^2}{2\sigma^2} + \log(\pi_k)$$

Hence:

$$\arg \max_k p_k(x) = \arg \max_k \delta_k(x)$$

Conclusion

Therefore, classifying an observation to the class for which (4.17) is largest is equivalent to classifying it to the class for which (4.18) is largest. Under the assumption that the observations in each class are normally distributed with a common variance σ^2 , the Bayes classifier assigns an observation to the class for which the discriminant function $\delta_k(x)$ is maximized.

2 Question 7

Problem Statement

Suppose that we wish to predict whether a given stock will issue a dividend this year (“Yes” or “No”) based on X , last year’s percent profit. The mean value of X for companies that issued a dividend was 10, while the mean for those that did not was 0. The variance for both groups was 36. Finally, 80% of companies issued dividends. Assuming that X follows a normal distribution, predict the probability that a company will issue a dividend given that its percentage profit was $X = 4$ last year.

Solution

We want to find the probability that a company issues a dividend given that its profit last year was $X = 4$:

$$P(\text{Yes} \mid X = 4)$$

Given Information

For companies that issued dividends:

$$\mu_{\text{Yes}} = 10$$

For companies that did not issue dividends:

$$\mu_{\text{No}} = 0$$

The variance for both groups is:

$$\sigma^2 = 36, \quad \sigma = 6$$

The prior probabilities are:

$$P(\text{Yes}) = 0.80, \quad P(\text{No}) = 0.20$$

Step 1: Apply Bayes’ Theorem

Bayes’ theorem gives:

$$P(\text{Yes} \mid X = 4) = \frac{P(\text{Yes}) f_{\text{Yes}}(4)}{P(\text{Yes}) f_{\text{Yes}}(4) + P(\text{No}) f_{\text{No}}(4)}$$

Therefore, we need to calculate the two normal densities.

Step 2: Calculate $f_{\text{Yes}}(4)$

The normal density is:

$$f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left[-\frac{(x - \mu)^2}{2\sigma^2}\right]$$

For the Yes group, $x = 4$, $\mu_{\text{Yes}} = 10$, and $\sigma^2 = 36$:

$$f_{\text{Yes}}(4) = \frac{1}{\sqrt{2\pi(36)}} \exp\left[-\frac{(4 - 10)^2}{2(36)}\right]$$

Since $(4 - 10)^2 = 36$:

$$f_{\text{Yes}}(4) = \frac{1}{6\sqrt{2\pi}} \exp(-36/72) = \frac{1}{6\sqrt{2\pi}} \exp(-0.5) \approx 0.0403$$

Step 3: Calculate $f_{\text{No}}(4)$

For the No group, $x = 4$, $\mu_{\text{No}} = 0$, and $\sigma^2 = 36$:

$$f_{\text{No}}(4) = \frac{1}{\sqrt{2\pi(36)}} \exp\left[-\frac{(4-0)^2}{2(36)}\right]$$

Since $(4-0)^2 = 16$:

$$f_{\text{No}}(4) = \frac{1}{6\sqrt{2\pi}} \exp(-16/72) = \frac{1}{6\sqrt{2\pi}} \exp(-0.2222) \approx 0.0532$$

Step 4: Apply Bayes' Theorem

Substituting the values:

$$P(\text{Yes} \mid X = 4) = \frac{0.8(0.0403)}{0.8(0.0403) + 0.2(0.0532)}$$

The numerator is:

$$0.8(0.0403) = 0.03224$$

The denominator is:

$$0.03224 + 0.01064 = 0.04288$$

Therefore:

$$P(\text{Yes} \mid X = 4) = \frac{0.03224}{0.04288} \approx 0.7519$$

Final Answer

$$P(\text{Dividend} = \text{Yes} \mid X = 4) \approx 75.2\%$$

Interpretation: If a company's profit was 4% last year, the probability that it will issue a dividend this year is approximately 75.2%.

3 Question 12

Problem Statement

Suppose that we wish to classify an observation $X \in \mathbb{R}$ into apples and oranges. A logistic regression model is compared with the softmax formulation in equation (4.13).

(a) Log Odds in the Logistic Regression Model

The odds of orange versus apple are:

$$\frac{P(\text{orange} \mid X = x)}{P(\text{apple} \mid X = x)} = \exp(\hat{\beta}_0 + \hat{\beta}_1 x)$$

Therefore, the log odds are:

$$\log(\text{odds}) = \hat{\beta}_0 + \hat{\beta}_1 x$$

(b) Log Odds in the Softmax Model

Using the softmax formulation, the odds of orange versus apple are:

$$\frac{P(\text{orange} \mid X = x)}{P(\text{apple} \mid X = x)} = \exp\left[(\hat{\beta}_{\text{orange},0} - \hat{\beta}_{\text{apple},0}) + (\hat{\beta}_{\text{orange},1} - \hat{\beta}_{\text{apple},1})x\right]$$

Thus, the log odds are:

$$\log(\text{odds}) = (\hat{\beta}_{\text{orange},0} - \hat{\beta}_{\text{apple},0}) + (\hat{\beta}_{\text{orange},1} - \hat{\beta}_{\text{apple},1})x$$

(c) Given $\hat{\beta}_0 = 2$ and $\hat{\beta}_1 = -1$

We need the softmax coefficients to satisfy:

$$\hat{\beta}_{\text{orange},0} - \hat{\beta}_{\text{apple},0} = 2, \quad \hat{\beta}_{\text{orange},1} - \hat{\beta}_{\text{apple},1} = -1$$

One valid choice is:

$$\hat{\beta}_{\text{orange},0} = 2, \quad \hat{\beta}_{\text{orange},1} = -1, \quad \hat{\beta}_{\text{apple},0} = 0, \quad \hat{\beta}_{\text{apple},1} = 0$$

There are infinitely many valid choices because adding the same value to both class coefficients does not change the probabilities.

(d) Convert the Friend's Coefficients to the Logistic Model

Given:

$$\hat{\beta}_{\text{orange},0} = 1.2, \quad \hat{\beta}_{\text{orange},1} = -2, \quad \hat{\beta}_{\text{apple},0} = 3, \quad \hat{\beta}_{\text{apple},1} = 0.6$$

The logistic intercept is:

$$\hat{\beta}_0 = 1.2 - 3 = -1.8$$

The logistic slope is:

$$\hat{\beta}_1 = -2 - 0.6 = -2.6$$

Therefore:

$$P(\text{orange} \mid X = x) = \frac{\exp(-1.8 - 2.6x)}{1 + \exp(-1.8 - 2.6x)}$$

(e) Expected Agreement

Both models produce exactly the same log odds, $-1.8 - 2.6x$. Therefore, they produce the same probabilities and the same predicted class labels.

For 2,000 test observations, the expected agreement is:

100% agreement, or 2,000 out of 2,000 predictions.

Final Answers Summary

Part	Answer
(a)	$\hat{\beta}_0 + \hat{\beta}_1 x$
(b)	$(\hat{\beta}_{\text{orange},0} - \hat{\beta}_{\text{apple},0}) + (\hat{\beta}_{\text{orange},1} - \hat{\beta}_{\text{apple},1})x$
(c)	One valid set: orange $(2, -1)$, apple $(0, 0)$
(d)	$\hat{\beta}_0 = -1.8, \hat{\beta}_1 = -2.6$
(e)	100% agreement
